1D "Warm Up" - Approximation Theory

Aim: Illustrate key ideas about model accuracy in "simple" 1D context.

Problem 2. Polynomial Regression

Given samples from a continuous map,

(*) $$y_i = G(x_i) \quad \text{for } i = 0, \ldots, n,$$

where $G : [0,1] \to \mathbb{R}$, find a degree $m \leq n$ polynomial that "best fits" the data,

(**) $$p_m = \underset{p \in \mathbb{P}_m}{\operatorname{argmin}} \, \| Y - p(X) \|,$$

where $Y = [y_0, \ldots, y_n]^T$ and $X = [x_0, \ldots, x_n]^T$.

$\Rightarrow$ Focus on $\|X\| = \sqrt{\sum_{k=0}^{n} x_i^2}$ today, corresponding to "least-squares" fit.

Algorithm 2. "Dictionary" Regression

1. Select basis $\{e_0(x), \ldots, e_m(x)\} \in \mathbb{P}_m$.

2. Minimize objective function

$$R(C) = \left\| \begin{bmatrix} e_0(x_0) & \cdots & e_m(x_0) \\ \vdots & & \vdots \\ e_0(x_n) & & e_m(x_n) \end{bmatrix} \begin{bmatrix} c_0 \\ \vdots \\ c_m \end{bmatrix} - \begin{bmatrix} y_0 \\ \vdots \\ y_n \end{bmatrix} \right\|$$

where $C = [c_0, \ldots, c_m]^T$.

3. The best degree $\leq m$ polynomial is

$$p_m(x) = c_0 e_0(x) + \cdots + c_m e_m(x).$$

Q: How accurate is the "fit"?

"Model Error" $$E_{n,m} = \sup_{0 \leq x \leq 1} |G(x) - p_m(x)|$$

Q: Given n, how do choose m?

⇒ See demo02.m for numerical experiments w/polynomial regression.

⇒ Remarkably, when $n \gg m$, the model error closely tracks the best $\mathbb{P}_m$ approximant – similar to interpolation at Chebyshev nodes.

Analysis of Regression. (View 1)

Consider a "continuous" analogue of Step 2 in Algorithm 2:

Discrete least-squares

$$\underset{c_0,\dots,c_m}{\text{minimize}} \sum_{j=0}^{n} \left[y_j - \sum_{k=0}^{m} c_k e_k(x_j) \right]^2$$

continuous least-squares

$$\underset{c_0,\dots,c_m}{\text{minimize}} \overbrace{\int_0^1 \left[G(x) - \sum_{k=0}^{m} c_k e_k(x) \right]^2 dx}^{\text{"}L^2\text{-norm" of } G - p_m}$$

⇒ The minimizer of the continuous problem is the best degree n polynomial approx. to G when measured in the $L^2[0,1]$ norm.

⇒ Just like the $C[0,1]$ problem, the minimizer exists, is unique, and converges algebraically with power dependent on smoothness of G.

⇒ The discrete objective is essentially a quadrature approximation of the continuous objective. As $n \to \infty$ (m fixed), the quadrature is "refined" and the minimizer of the discrete problem looks increasingly like the minimizer of the continuous problem!

If data is equispaced with $h = |x_{j+1} - x_j|$,

$$\underbrace{h \sum_{j=0}^{n} \left[y_j - \sum_{k=0}^{m} c_k e_k(x_j) \right]^2}_{\text{Riemann Sum}} \xrightarrow{n \to \infty} \int_0^1 [G(x) - p(x)]^2 \, dx$$

If data is not equispaced, $h_j = |x_{j+1} - x_j|$,

$$\underbrace{\sum_{j=0}^{n} h_j \Big[y_j - \sum_{k=0}^{m} c_k e_k(x_j)\Big]^2}_{\text{Riemann Sum}} \xrightarrow{n\to\infty} \int_0^1 [G(x) - p(x)]^2 dx$$

provided that $\limsup_{j\to\infty} h_j = 0$.

The modified objective for the discrete problem leads to weighted least-squares.

Analysis of Regression. (View 2)

The sensitivity of Algorithm 2 outputs to small perturbations in the inputs is measured by the condition #

$$K(A) = \frac{\sigma_1(A)}{\sigma_m(A)}, \qquad A = \begin{bmatrix} e_0(x_0) & \cdots & e_m(x_0) \\ \vdots & & \vdots \\ e_0(x_n) & \cdots & e_m(x_n) \end{bmatrix},$$

where $\sigma_1(A) \geq \cdots \geq \sigma_m(A)$ are the singular values of the matrix A.

Q: Why does oversampling ($m<n$) improve the condition # of A?

$\Rightarrow$ See demo02.m for num. experiments.

The singular values of A are related to the eigenvalues of the Gram matrix:

$$A^T A v_j = \sigma_j^2 v_j \qquad j = 0, \ldots, m.$$

What happens to the eigenvalues of $A^T A$?

Idea 1. Convergence to L^2 Gram matrix.

The entries of the Gram matrix are

$$\frac{1}{n}(A^T A)_{ij} = \frac{1}{n}\sum_{k=0}^{n} e_i(x_k)\, e_j(x_k)$$

If $\sup_{1\le k\le n} |x_k - x_{k-1}| \to 0$ as $n\to\infty$, then

$$\lim_{n\to\infty} \frac{1}{n}(A^T A)_{ij} = \int_0^1 e_i(x) e_j(x)\, dx$$

So the (re-scaled) Gram matrices converge to the L^2 Gram matrix associated w/ the basis $\{e_0, \ldots, e_m\}$!

$\Rightarrow$ "Good" basis (ONB) has

$$\lim_{n\to\infty} \frac{1}{n}(A^T A)_{ij} = \begin{cases} 1 & i=j \\ 0 & i\neq j \end{cases}$$

which has condition number $= \underline{1}$!

$\Rightarrow$ "Bad basis (e.g. monomials) has large condition # for L^2 Gram matrix.

$$\lim_{n\to\infty} \frac{1}{n}(A^T A)_{ij} = \int_0^1 x^{i+j}\,dx$$

$$= \frac{2}{1+i+j} \qquad 0\leq i,j\leq m$$

This is a real Cauchy matrix, whose condition # grows exp. as m increases.

Idea 2. Eigenvalues of matrices depend continuously on the matrix entries.

So, if $A_n \to A_*$ as $n \to \infty$, then

$$\lambda_j(A_n) \to \lambda_j(A) \text{ as } n \to \infty,$$

and, in particular, $K(A_n) \to K(A)$.

(If $\sigma_{min}(A) = 0$, then $K(A_n) \to \infty$.)

Conclusion: If m is fixed and $\{e_0, \dots, e_m\}$ is a "good" basis, $K(A_n)$ should approach a modest constant as n increases, determined by

$$K(E), \quad \text{where} \quad E_{ij} = \int_0^1 e_i(x) e_j(x) dx,$$

i.e., the L^2 Gram matrix of the basis.

Limitations: This argument is about refining n while m remains fixed.

$\Rightarrow$ For a full understanding of "over-sampling," i.e., refining $m = h(n)$ as $n \to \infty$, a much more careful analysis of the Gram matrices is required.

Big Picture: "Oversampling" to fit a polynomial to data is a form of regularization.

$\Rightarrow$ Regularization trades model accuracy for "stability," or more precisely, a better condition # for the underlying mathematical problem.

$\Rightarrow$ Condition #s describe sensitivity of model to perturbed data, i.e., noisy observations, etc.